

Docker Commands Cheat Sheet

Using nginx as example

1) Check Docker Installation

Check Docker version:

```
docker --version
```

Detailed client/server version:

```
docker version
```

Check whether Docker is working:

```
docker info
```

Check Docker service status (Linux):

```
systemctl status docker
```

Start Docker service:

```
sudo systemctl start docker
```

Enable Docker on boot:

```
sudo systemctl enable docker
```

2) Get Help

General help:

```
docker --help
```

Help for specific command:

```
docker run --help
```

```
docker image --help
```

3) Work with Images

List all local images:

```
docker images
```

```
docker image ls
```

Search images on Docker Hub:

```
docker search nginx
```

Pull the latest nginx image:

```
docker pull nginx
```

Pull a specific version:

```
docker pull nginx:1.25
```

Remove an image:

```
docker rmi nginx
```

Inspect image details:

```
docker image inspect nginx
```

Show image history:

```
docker history nginx
```

4) Work with Containers

List running containers:

```
docker ps
```

List all containers (including stopped):

```
docker ps -a  
docker container ls -a
```

Run nginx in detached mode:

```
docker run -d nginx
```

Run nginx with custom name:

```
docker run -d --name mynginx nginx
```

Run nginx with port mapping (8080→80):

```
docker run -d --name mynginx -p 8080:80 nginx
```

Run nginx interactively:

```
docker run -it nginx sh
```

Run and auto-remove after exit:

```
docker run --rm nginx
```

Start a stopped container:

```
docker start mynginx
```

Stop a running container:

```
docker stop mynginx
```

Restart a container:

```
docker restart mynginx
```

Pause/Unpause a container:

```
docker pause mynginx  
docker unpause mynginx
```

Kill a container immediately:

```
docker kill mynginx
```

Remove a stopped container:

```
docker rm mynginx
```

Force remove running container:

```
docker rm -f mynginx
```

Rename a container:

```
docker rename mynginx webserver
```

5) Inspect and Monitor Containers

View logs:

```
docker logs mynginx
```

Follow logs live:

```
docker logs -f mynginx
```

Show running processes:

```
docker top mynginx
```

Inspect container details:

```
docker inspect mynginx
```

Show resource usage:

```
docker stats
docker stats mynginx
```

See port mappings:

```
docker port mynginx
```

6) Enter a Running Container

Open shell inside container:

```
docker exec -it mynginx sh
docker exec -it mynginx bash
```

Run single command:

```
docker exec mynginx ls /usr/share/nginx/html
```

7) Copy Files Between Host and Container

Copy from host to container:

```
docker cp index.html mynginx:/usr/share/nginx/html/
```

Copy from container to host:

```
docker cp mynginx:/etc/nginx/nginx.conf .
```

8) Modify Images and Containers

Save changes as new image:

```
docker commit mynginx custom-nginx:v1
```

Tag an image:

```
docker tag custom-nginx:v1 myrepo/custom-nginx:v1
```

Build from Dockerfile:

```
docker build -t custom-nginx:v1 .
```

9) Create Images Properly

Build image from Dockerfile:

```
docker build -t custom-nginx:v1 .
```

Build with specific Dockerfile:

```
docker build -t custom-nginx:v1 -f Dockerfile .
```

Build without cache:

```
docker build --no-cache -t custom-nginx:v1 .
```

10) Save, Load, Export, Import

Save image to tar:

```
docker save -o nginx-image.tar nginx
```

Load image from tar:

```
docker load -i nginx-image.tar
```

Export container filesystem:

```
docker export mynginx -o mynginx-container.tar
```

Import as image:

```
docker import mynginx-container.tar imported-nginx:v1
```

11) Networks

List networks:

```
docker network ls
```

Inspect network:

```
docker network inspect bridge
```

Create custom network:

```
docker network create mynetwork
```

Run on custom network:

```
docker run -d --name mynginx --network mynetwork -p 8080:80 nginx
```

Connect to network:

```
docker network connect mynetwork mynginx
```

Disconnect from network:

```
docker network disconnect mynetwork mynginx
```

Remove network:

```
docker network rm mynetwork
```

12) Volumes

List volumes:

```
docker volume ls
```

Create volume:

```
docker volume create nginx-data
```

Run with volume:

```
docker run -d --name mynginx -p 8080:80 -v nginx-data:/usr/share/nginx/html nginx
```

Mount current folder:

```
docker run -d --name mynginx -p 8080:80 -v $(pwd):/usr/share/nginx/html nginx
```

Inspect volume:

```
docker volume inspect nginx-data
```

Remove volume:

```
docker volume rm nginx-data
```

13) Cleanup Commands

Remove stopped containers:

```
docker container prune
```

Remove unused images:

```
docker image prune
```

Remove all unused images:

```
docker image prune -a
```

Remove unused volumes:

```
docker volume prune
```

Remove unused networks:

```
docker network prune
```

Remove everything unused:

```
docker system prune
```

Remove everything (including images):

```
docker system prune -a
```

14) Useful One-Liners

Stop all running containers:

```
docker stop $(docker ps -q)
```

Remove all containers:

```
docker rm $(docker ps -aq)
```

Remove all images:

```
docker rmi $(docker images -q)
```

15) Docker Compose Basics

Start services:

```
docker compose up -d
```

View running services:

```
docker compose ps
```

View logs:

```
docker compose logs -f
```

Stop and remove services:

```
docker compose down
```

16) Most-Used Beginner Workflow

```
docker --version
docker info
docker pull nginx
docker images
docker run -d --name mynginx -p 8080:80 nginx
docker ps
docker logs mynginx
docker exec -it mynginx sh
docker stop mynginx
docker start mynginx
docker rm -f mynginx
docker rmi nginx
```

17) Quick Notes

- You run containers, not images
- Images are templates; containers are running instances
- Stop/start/restart/remove containers
- Remove images, but don't 'stop' images
- Use Dockerfile for permanent changes

Mini Example - Quick Start

```
docker pull nginx
docker run -d --name mynginx -p 8080:80 nginx
```

Then open: <http://localhost:8080>